

Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

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Eligibility Criteria

Healthy participants, as determined by medical history, physical examination, and clinical judgment, were eligible for inclusion in the study. Participants were receiving prenatal standard of care based on country requirements; had a fetal anomaly ultrasound examination performed at ≥ 18 weeks of pregnancy with no significant abnormalities observed; and had a documented negative HIV antibody test, syphilis test, and hepatitis B virus surface antigen test during the current pregnancy and before randomization.

Participants were excluded if they had bleeding diathesis or a condition associated with prolonged bleeding; a history of severe adverse reaction associated with a vaccine and/or severe allergic reaction (eg, anaphylaxis) to any component of the product or any related vaccine; or a major illness of the maternal participant or conditions of the fetus that, in the investigator's judgment, would substantially increase the risk associated with the maternal or infant participant's participation in and completion of the study or could preclude the evaluation of the maternal participant's response (includes positive serologic testing for regional endemic conditions assessed during routine maternal care, as per local standards of care and obstetric recommendations).

Exclusion criteria related to the current pregnancy included in vitro fertilization; pregnancy complications or abnormalities at the time of consent that would increase the risk associated with participation in and completion of the study (including but not limited to preeclampsia, eclampsia, or uncontrolled gestational hypertension; placental abnormality; polyhydramnios or oligohydramnios; significant bleeding or a blood clotting disorder; endocrine disorders, including untreated hyperthyroidism or untreated hypothyroidism or disorders of glucose intolerance antedating pregnancy or occurring during pregnancy if uncontrolled at the time of consent; any signs of premature labor with the pregnancy or having ongoing intervention [medical/surgical] to prevent preterm birth).

Exclusion criteria related to prior pregnancies included those that would increase the risk associated with the participation in and completion of the study (including but not limited to prior preterm delivery at ≤ 34 weeks' gestation; prior stillbirth or neonatal death; previous infant with a known genetic disorder or significant congenital anomaly).

Other exclusion criteria for maternal participants were congenital or acquired immunodeficiency disorder, or rheumatologic disorder or other illness requiring chronic treatment with known immunosuppressant medications, including monoclonal antibodies, within the year before enrolment or other acute or chronic medical or psychiatric condition including recent (within the past year) or active suicidal ideation or behavior or laboratory abnormality that may have increased the risk

associated with study participation or investigational product administration or with the interpretation of study results.

Maternal participants were also excluded if they received the following prior or concomitant therapies: investigational drug(s) within 28 days before consent and/or during study participation; monoclonal antibodies within the year before enrolment or systemic corticosteroids for >14 days within 28 days before study enrolment (participants could receive SARS-CoV-2 monoclonal antibodies, prednisone doses of <20 mg/day for ≤14 days and inhaled/nebulized, intra-articular, intrabursal, or topical corticosteroids). Other exclusion criteria included alcohol abuse or illicit drug use and receipt of blood or plasma products or immunoglobulin, from 60 days before investigational product administration, or planned receipt through delivery (except Rho[D] immune globulin). Participants could not have received any licensed or investigational RSV vaccine or planned receipt during study participation.

Ethical Conduct of the Study

The study was conducted in compliance with ethical principles of the Declaration of Helsinki for medical research involving human subjects, the Council for International Organizations of Medical Sciences International Ethical Guidelines, the International Council for Harmonization Good Clinical Practice guidelines, and applicable laws and regulations. The protocol, protocol amendments, informed consent document, and other relevant documents were reviewed and approved by the sponsor and by the institutional review board/ethics committee before study initiation. All participants provided informed consent; this included consent for the mother and fetus during the pregnancy and the infant's continuation in the study after delivery.

Statistical Considerations

Sample Size Determination and Hypothesis Testing

This was an event-driven study. Sample size was determined as the number of participants randomized expected to accrue the required number of endpoint cases in the evaluable population. Success on either of the two primary endpoints of the current study was sufficient. Therefore, power for the study overall was at least as high as the power for either primary endpoint individually. For the study to have ≥90% power to reject the null hypothesis for RSV-MA-LRTI when true VE was 60%, a total of 124 cases of RSV-MA-LRTI within 90 days of birth were required in the evaluable population. This also accounted for potential use of an alpha of 1.25% one-sided within the multiplicity adjustment. There was no explicit case target for the endpoint of severe RSV-MA-LRTI.

Depending on the assumed true VE, power for that individual hypothesis may have been lower, but the power for the primary endpoint family was $\geq 90\%$.

This global study was conducted in the United States, other high-income countries, and lower- and middle-income countries. Incidence rates of the primary endpoints were expected to vary in different regions. Based on a previous phase 3 trial of an RSV vaccine,¹ the assumed incidence rate for RSV-MA-LRTI through 90 days in low-incidence countries, such as the United States, was approximately 1.75%; and in other regions was approximately 3.90%. With these assumptions and allowing for 60% VE and 10% of enrolled participants being nonevaluable, enrollment of approximately 6900 participants was planned for the study as a whole. The exact number of participants required to accrue the case target could be higher or lower than this, depending on incidence rates, observed VE, and the proportion of evaluable participants.

Interim Analysis

The analysis of efficacy used an O'Brien-Fleming alpha spending rule based on the fraction of cases of RSV-MA-LRTI within the 90 days available. The exact number of cases at each interim analysis was not fixed and could be decided based on operational reasons. However, no fewer than 43 cases were to be included in the first interim analysis, and no fewer than 62 cases were to be included in the second interim analysis. The alpha levels used at the interim and final analyses depended on the exact fraction of cases available at the interim analysis. For the endpoints inspected at the interim analysis (MA-LRTI and severe MA-LRTI at 90 days), the O'Brien-Fleming alpha spending function determined the alpha to be used, split between the endpoints using the Bonferroni correction. Because MA-LRTI through 180 days was inspected at the first interim analysis with a 2-sided alpha of 0.0017, the analysis of the primary endpoints at 120 days and later after a successful interim analysis used a 2-sided alpha of $0.05 - 0.0017 = 0.0483$, split between the endpoints using the Bonferroni correction.

First interim analysis

The first interim analysis was conducted in April 2022 when 56 evaluable cases of MA-LRTI due to RSV within 90 days after birth had accrued in infant participants. The efficacy criterion (lower bound of CI > 20%) was met for cases within 90 days after birth but not within 150 days. The external Data Monitoring Committee reviewed the results and recommended continuation of the study. The double-blind was maintained after the first interim analysis.

¹ Madhi et al. N Engl J Med. 2020;383:426-439.

Statistical Analyses of the Primary Efficacy Endpoints

The analysis of efficacy used a conditional exact test based on the binomial distribution of the number of disease cases in the RSVpreF group, given the total number of cases in both groups. The infant evaluable efficacy population was the primary population for efficacy analyses (defined in **Table S2**). The two primary endpoints of severe RSV-MA-LRTI and RSV-MA-LRTI were tested in parallel using a Bonferroni adjustment. Success of the trial required rejection of the null hypothesis (ie, a CI lower bound >20%) for either of the two primary endpoints.

Testing of the two primary endpoints across the time intervals followed a fixed sequence with a gatekeeping strategy. Only primary endpoint(s) that were successful at all earlier time intervals were considered at subsequent intervals.

Confidence intervals for VE were calculated using the appropriate multiplicity-adjusted alpha level α . If the lower $100(1-\alpha)\%$ confidence limit for VE exceeded 20%, the null hypothesis for that endpoint was rejected. Accrual of endpoint cases over 180 days are presented using Kaplan–Meier plots.

If the hypothesis for both primary endpoints was successful through 180 days, the three secondary endpoints would be analyzed using $\alpha = 0.05/3$ assigned to each; if the hypothesis for one primary endpoint was not successful through 180 days, the secondary endpoints would be analyzed using $\alpha = 0.025/3$ assigned to each. Testing of each secondary endpoint across the time intervals followed a fixed sequence beginning with the first interval, using the full alpha allocated to the endpoint. Testing at subsequent intervals required that the endpoint was successful at all earlier time intervals. For all secondary efficacy endpoints, RSVpreF was compared with placebo, testing the hypotheses $H_0: VE \leq 0\%$ versus $H_a: VE > 0\%$ (ie, a CI lower bound >0%). Hypothesis testing of the secondary endpoints is conditional upon rejection of the null hypothesis for ≥ 1 of the primary endpoints.

Sensitivity Analyses of the Primary Efficacy Endpoints

A sensitivity analysis of the primary endpoints examined the impact of missing RSV swab results. Cases that qualified as MA-LRTI events based on clinical signs and symptoms but with no valid RSV swab result were identified. Sensitivity analyses of the primary endpoints were conducted under various assumptions about the missing swab results, from a missing-at-random scenario and several missing-not-at-random scenarios unfavorable to the vaccine. In each scenario, cases with imputed positive RSV results were added to the confirmed RSV-positive cases and vaccine efficacy in the augmented set of cases analyzed.

Multiple imputations were performed to randomly assign missing swab results. SAS PROC MI (SAS Institute, Cary, NC, USA) was used to generate 500 imputed data sets for each scenario. Mean and median VE across imputations, and the proportion of imputations with a VE lower bound CI >20% are reported. Even in the situation in which close to 100% of missing results in the vaccine group are assumed to be positive, 100% of the imputed datasets for severe MA-LRTI resulted in CI lower bounds >20% at all timepoints.

Table S1. Case definitions used for infant participants in the study

Case	Definition
MA visit	Infant participant has been taken to or seen by a healthcare provider (eg, outpatient, inpatient visit, emergency room, urgent care, or home visit)
MA-RTI visit	A medically-attended visit AND ≥ 1 of the following RTI signs and symptoms: - Nasal discharge for ≥ 24 hours - Difficulty breathing, labored breathing, or rapid breathing (any duration) - Cough - Inability to feed for any duration because of respiratory symptoms - Apnea - Any other respiratory symptom of concern
RSV-MA-RTI	An MA-RTI visit AND RSV-positive test result*
MA-LRTI due to any cause	MA-RTI visit AND ≥ 1 of the following: - Fast breathing (RR ≥ 60 bpm for <2 months of age [<60 days of age], ≥ 50 bpm for 2–<12 months of age, or ≥ 40 bpm for 12–24 months of age) - SpO₂ $<95\%$ - Chest wall indrawing
RSV-MA-LRTI	MA-RTI visit AND RSV-positive test result* AND ≥ 1 of the following - Fast breathing (RR ≥ 60 bpm for <2 months of age [<60 days of age], ≥ 50 bpm for 2–<12 months of age, or ≥ 40 bpm for 12–24 months of age) - SpO₂ $<95\%$ - Chest wall indrawing
Hospitalized RSV-RTI†	RTI due to RSV that results in hospitalization
Severe RSV-MA-LRTI†	MA-RTI visit AND RSV-positive test result* AND ≥ 1 of the following - Fast breathing (RR ≥ 70 bpm for <2 months of age [<60 days of age], ≥ 60 bpm for 2–<12 months of age, or ≥ 50 bpm for 12–24 months of age) - SpO₂ $<93\%$ - High-flow nasal cannula or mechanical ventilation (ie, invasive or noninvasive) - ICU admission for >4 hours - Failure to respond/unconscious

bpm=breaths per minute; ICU=intensive care unit; LRTI=lower respiratory tract illness; MA=medically-attended; NAAT=nucleic acid amplification technology; RR=respiratory rate; RSV=respiratory syncytial virus; RTI=respiratory tract illness; RT-PCR=reverse transcription-polymerase chain reaction; SpO₂=oxygen saturation.

* RSV RT-PCR–positive test result by Pfizer central laboratory or by certified laboratory with NAAT for RSV, which was conducted on a sample obtained during the medically-attended visit or within 10 days (where Day 1 is the day of the MA-RTI visit).

† An endpoint adjudication committee determined if the endpoint criteria were met upon review of the site source documentation from the MA-RTI visit and RTI study visits, including all available RSV test results.

Table S2. Study populations

Population	Description
Evaluable efficacy – infant (per protocol)	All infant participants who were eligible, were born to the maternal participants who received the investigational product to which they were randomized at least 14 days before delivery, did not receive palivizumab or another monoclonal antibody targeting RSV, had no major protocol violations, and did not have transfusions of more than 20 mL/kg of any blood products at <180 days. • Includes 3495 infants born to RSVpreF maternal recipients • Includes 3480 infants born to placebo maternal recipients.
Safety – infant	All infant participants who were born to vaccinated maternal participants. • Includes 3568 infants* born to RSVpreF maternal recipients • Includes all 3558 infants born to placebo maternal recipients.
Safety – maternal	All randomized maternal participants who received investigational product. • Includes all 3682 RSVpreF maternal recipients • Includes 3675†placebo maternal recipients.

RSV=respiratory syncytial virus; RSVpreF=respiratory syncytial virus prefusion F protein–based vaccine.

* Two enrolled infant participants did not meet eligibility criteria and were not included in the safety population.

† One maternal participant who received placebo did not meet eligibility criteria and was not included in the safety population.

Table S3. Representativeness of study participants

Category	Example
Disease, problem, or condition under investigation	RSV-related illness
Special considerations related to	
Sex	Male sex is a risk factor for severe RSV-LRTI in children.
Age	RSV-related illness is most prevalent in infants and young children, followed by older adults and high-risk adults.
Race or ethnic group	Native American and Alaska Native infants and children, infants of Māori or Pacific ethnicity, and children of Black race have higher RSV-related hospitalization rates compared with other ethnicities and races.
Geography	RSV has a substantial global healthcare burden. In temperate climates, RSV causes yearly epidemics over winter months generally lasting 3–4 months. In tropical climates, no distinct seasonality is apparent and outbreaks are more unpredictable and continuous.
Other considerations	Risk factors for severe disease in infants include: • Premature birth • Bronchopulmonary dysplasia • Congenital heart disease • Neuromuscular impairment • Immunodeficiency • Down syndrome • <6 months old at the beginning of RSV season • Multiple birth • Male sex • Low socioeconomic status
Overall representativeness of this study	The infants included in this study were generally representative of a population at risk of RSV-related illness. Overall, 51% of infant participants were males, 64% were White, 19% were Black, and 12% were Asian; only 1.1% and 0.3% were Native American/Alaskan Native or Native Hawaiian or other Pacific Islander, respectively. The gestational age at birth was 24–<37 weeks in 5% of infant participants. Participants were enrolled from Argentina, Australia, Brazil, Canada, Chile, Denmark, Finland, Gambia, Japan, Mexico, Netherlands, New Zealand, Philippines, Republic of Korea, South Africa, Spain, Taiwan, and the United States. The countries with the highest enrollment were the United States (45.4% of maternal participants), South Africa (13.0%), Argentina (12.4%), and Japan (6.3%); other countries enrolled <5% of participants each.

LRTI=lower respiratory tract illness; RSV=respiratory syncytial virus.

Table S4. Number and percentage of maternal participants randomized by country

Country	Maternal participants randomized, n (%)
United States	3353 (45.4)
South Africa	964 (13.0)
Argentina	914 (12.4)
Japan	462 (6.3)
Taiwan	260 (3.5)
Spain	246 (3.3)
Gambia	196 (2.7)
Netherlands	194 (2.6)
Chile	171 (2.3)
Finland	150 (2.0)
New Zealand	100 (1.4)
Philippines	80 (1.1)
Mexico	75 (1.0)
Brazil	74 (1.0)
Denmark	62 (0.8)
Canada	55 (0.7)
Australia	24 (0.3)
Republic of Korea	12 (0.2)

Table S5. Sensitivity analysis for missing laboratory results for severe RSV-MA-LRTI occurring within 90 to 180 days after birth

Time interval	Imputed value positive rate across all imputations [RSVpreF:placebo]* (%)	Infection rates based on existing and imputed values [RSVpreF:placebo]† (%)	Number of VE (100-α _‡)% CI LB>20%	Percentage of VE (100-α _‡)% CI LB>20%	Average VE (%)
90 days after birth	38.0:56.8§	107.7:499.0	500	100	81.5
	44.3:56.8	114.7:499.0	500	100	80.3
	55.4:56.8	124.9:499.0	500	100	78.6
	86.0:56.8	145.1:499.0	500	100	75.1
	99.7:56.8	152.4:499.0	500	100	73.9
120 days after birth	33.0:56.2§	154.3:484.2	500	100	73.3
	43.0:56.2	163.0:484.2	500	100	71.8
	60.3:56.2	174.4:484.2	500	100	69.9
	90.8:56.2	190.6:484.2	500	100	67.1
	99.7:56.2	195.0:484.2	500	100	66.3
150 days after birth	26.1:51.7§	169.0:478.5	500	100	70.0
	37.8:51.7	179.3:478.5	500	100	68.2
	60.6:51.7	194.1:478.5	500	100	65.6
	91.7:51.7	212.9:478.5	500	100	62.2
	99.8:51.7	217.7:478.5	500	100	61.4
180 days after birth	23.5:50.3§	163.9:449.4	500	100	68.4
	38.2:50.3	174.7:449.4	500	100	66.3
	60.1:50.3	188.6:449.4	500	100	63.6
	91.3:50.3	206.4:449.4	500	100	60.2
	99.8:50.3	211.3:449.4	500	100	59.3

Data are for the evaluable efficacy population (**Table S2**).

LB=lower bound; MA-LRTI=medically-attended lower respiratory tract illness; RSV=respiratory syncytial virus; RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine; VE=vaccine efficacy.

Each row of this table represents 500 imputations runs generated using SAS PROC MI. A logistic regression model was built of the form, primary endpoint response = vaccine group. Based on complete data before imputation, positive rates (number of positive cases divided by number of subjects with swabs and results available) were (0.13, 0.14, 0.15 ,0.15) and (0.3, 0.31, 0.3, 0.31) for 90/120/150/180 days after birth for the RSVpreF and placebo groups, respectively. Within a row, each imputation filled in the missing results with plausible results based on the logistic regression model at the subject level.

Cases were confirmed by an endpoint adjudication committee.

* Overall positive rates for the two vaccine groups based only on the imputed result data and across all the imputations within a row. Under the missing at random (MAR) assumption, the imputation model projects the probability of positive cases for the RSVpreF and placebo vaccine groups based on the complete data. Bias was applied to the model to increase the positive rate for the imputed values for the RSVpreF group only, to mimic potential and unknowable not at random (MNAR) effects that are unfavorable for efficacy conclusions for the study.

† Infection rates are number of positive cases per 1000 subjects within RSVpreF and placebo groups respectively.

‡ α = two-sided 0.5% for 90 days after birth and two-sided 2.48% for 120/150/180 days after birth.

§ This row represents the missing at random (MAR) assumption to which no bias has been applied to the model to increase the positive rate for the imputed RSVpreF group.

Table S6. Vaccine efficacy for RSV-MA-LRTI within 210 to 360 days after birth

	RSVpreF (N*=3495)	Placebo (N†=3480)	Vaccine efficacy, % (99.17% CI)†
Time interval	Number of cases (%)	Number of cases (%)	
210 days after birth	70 (2.0)	127 (3.6)	44.9 (17.9–63.5)
240 days after birth	76 (2.2)	133 (3.8)	42.9 (16.1–61.6)
270 days after birth	82 (2.3)	137 (3.9)	40.1 (13.0–59.2)
360 days after birth	92 (2.6)	156 (4.5)	41.0 (16.2–58.9)

Data are for the evaluable efficacy population (**Table S2**).

RSV-MA-LRTI=medically-attended lower respiratory tract illness due to respiratory syncytial virus;

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

* Number of participants at risk in the specified group. These values were used as denominators for percentage calculations.

† Vaccine efficacy was calculated as $1 - (P/[1 - P])$, where P is the number of cases in the RSVpreF group divided by the total number of cases. The confidence interval was adjusted using the Bonferroni procedure and accounting for the primary endpoints results. As a secondary endpoint, the criterion for vaccine efficacy was a lower bound of the confidence interval >0%.

Table S7. Hospitalizations due to RSV occurring within 90 to 360 days after birth

Time interval	RSVpreF (N*=3495) Number of cases (%)	Placebo (N*=3480) Number of cases (%)	Vaccine efficacy, % (99.17% CI)†
90 days after birth	10 (0.3)	31 (0.9)	67.7 (15.9–89.5)
120 days after birth	15 (0.4)	37 (1.1)	59.5 (8.3–83.7)
150 days after birth	17 (0.5)	39 (1.1)	56.4 (5.2–81.5)
180 days after birth	19 (0.5)	44 (1.3)	56.8 (10.1–80.7)
360 days after birth	38 (1.1)	57 (1.6)	33.3 (–17.6–62.9)

Data are for the evaluable efficacy population (**Table S2**).

LRTI=lower respiratory tract illness; RSV=respiratory syncytial virus; RSVpreF=respiratory syncytial virus prefusion F protein–based vaccine.

Cases were confirmed by an endpoint adjudication committee.

All infants had RSV-confirmed hospitalized illness but not all met severe LRTI criteria.

* Number of participants at risk in the specified group. These values were used as denominators for percentage calculations.

† Vaccine efficacy was calculated as $1-(P/[1-P])$, where P is the number of cases in the RSVpreF group divided by the total number of cases. The confidence interval was adjusted using the Bonferroni procedure and accounting for the primary endpoints results. As a secondary endpoint, the criterion for vaccine efficacy was a lower bound of the confidence interval >0%.

Table S8. Vaccine efficacy for all-cause MA-LRTI within 90 to 360 days after birth

Time interval	RSVpreF (N*=3495) Number of cases (%)	Placebo (N*=3480) Number of cases (%)	Vaccine efficacy, % (99.17% CI)†
90 days after birth	186 (5.3)	200 (5.7)	7.0 (–22.3–29.3)
120 days after birth	261 (7.5)	278 (8.0)	6.1 (–18.3–25.5)
150 days after birth	331 (9.5)	349 (10.0)	5.2 (–16.5–22.8)
180 days after birth	392 (11.2)	402 (11.6)	2.5 (–17.9–19.4)
360 days after birth	504 (14.4)	531 (15.3)	5.1 (–12.1–19.6)

Data are for the evaluable efficacy population (**Table S2**).

MA-LRTI=medically-attended lower respiratory tract illness; RSVpreF=respiratory syncytial virus prefusion F protein–based vaccine.

* Number of participants at risk in the specified group. These values were used as denominators for percentage calculations.

† Vaccine efficacy was calculated as $1-(P/[1-P])$, where P is the number of cases in the RSVpreF group divided by the total number of cases. The confidence interval was adjusted using the Bonferroni procedure and accounting for the primary endpoints results. As a secondary endpoint, the criterion for vaccine efficacy was a lower bound of the confidence interval >0%.

Table S9. Vaccine efficacy for RSV-MA-RTI within 90 to 180 days after birth

Time interval	RSVpreF (N*=3495) Number of cases (%)	Placebo (N*=3480) Number of cases (%)	Vaccine efficacy, % (95% CI)†
90 days after birth	67 (1.9)	110 (3.2)	39.1 (16.7–55.7)
120 days after birth	98 (2.8)	160 (4.6)	38.7 (20.8–52.9)
150 days after birth	126 (3.6)	209 (6.0)	39.7 (24.4–52.1)
180 days after birth	157 (4.5)	253 (7.3)	37.9 (24.0–49.5)

Data are for the evaluable efficacy population (**Table S2**).

RSV-MA-RTI=medically-attended respiratory tract illness due to respiratory syncytial virus;

RSVpreF=respiratory syncytial virus prefusion F protein–based vaccine.

Cases were confirmed by an endpoint adjudication committee.

* Number of participants at risk in the specified group. These values were used as denominators for percentage calculations.

† Vaccine efficacy was calculated as $1 - (P/[1 - P])$, where P is the number of cases in the RSVpreF group divided by the total number of cases. As an exploratory endpoint, no predefined criterion for vaccine efficacy was established. Confidence intervals should not be used to draw conclusions about these endpoints.

Table S10. Vaccine efficacy for RSV-MA-LRTI and severe RSV-MA-LRTI within 180 days after birth by RSV subgroup

Endpoint/RSV subgroup	RSVpreF (N*=3495) Number of cases (%)	Placebo (N*=3480) Number of cases (%)	Vaccine efficacy, % (95% CI)†
RSV-MA-LRTI			
RSV A	19 (0.5)	26 (0.7)	26.9 (–37.2–61.8)
RSV B	38 (1.1)	87 (2.5)	56.3 (35.4–71.0)
Severe DRSV-MA-LRTI			
RSV A	7 (0.2)	14 (0.4)	50.0 (–32.4–82.9)
RSV B	11 (0.3)	44 (1.3)	75.0 (50.8–88.4)

Data are for the evaluable efficacy population (**Table S2**).

RSV-MA-LRTI=medically-attended lower respiratory tract illness due to respiratory syncytial virus;

RSVpreF=respiratory syncytial virus prefusion F protein–based vaccine.

Cases were confirmed by an endpoint adjudication committee.

* Number of participants at risk in the specified group. These values were used as denominators for percentage calculations.

† Vaccine efficacy was calculated as $1 - (P/[1 - P])$, where P is the number of cases in the RSVpreF group divided by the total number of cases. As an exploratory endpoint, no predefined criterion for vaccine efficacy was established. Confidence intervals should not be used to draw conclusions about these endpoints.

Table S11. Adverse events* reported in maternal participants within 1 month of vaccination

System organ class Preferred term	RSVpreF (N†=3682) n‡ (%)	Placebo (N†=3675) n‡ (%)
Blood and lymphatic disorders	30 (0.8)	29 (0.8)
Anemia	14 (0.4)	22 (0.6)
Iron deficiency anemia	4 (0.1)	6 (0.2)
Cardiac disorders	21 (0.6)	20 (0.5)
Non-reassuring fetal heart rate pattern	5 (0.1)	7 (0.2)
Tachycardia	6 (0.2)	2 (<0.1)
Gastrointestinal disorders	34 (0.9)	36 (1.0)
Abdominal pain	7 (0.2)	3 (<0.1)
Nausea	6 (0.2)	2 (<0.1)
Vomiting	6 (0.2)	1 (<0.1)
General disorders and administration site conditions	17 (0.5)	20 (0.5)
Fatigue	1 (<0.1)	8 (0.2)
Hepatobiliary disorders	10 (0.3)	17 (0.5)
Cholestasis of pregnancy	5 (0.1)	13 (0.4)
Infections and infestations	72 (2.0)	75 (2.0)
Gastroenteritis	10 (0.3)	4 (0.1)
Upper respiratory tract infection	1 (<0.1)	6 (0.2)
Urinary tract infection	22 (0.6)	20 (0.5)
Vulvovaginal candidiasis	3 (<0.1)	7 (0.2)
Investigations	35 (1.0)	32 (0.9)
SARS-CoV-2 test positive	23 (0.6)	22 (0.6)
Musculoskeletal and connective tissue disorders	21 (0.6)	14 (0.4)
Back pain	6 (0.2)	8 (0.2)
Nervous system disorders	22 (0.6)	18 (0.5)
Headache	7 (0.2)	8 (0.2)
Pregnancy, puerperium, and perinatal conditions	256 (7.0)	228 (6.2)
Arrested labor	7 (0.2)	8 (0.2)
Fetal distress syndrome	9 (0.2)	8 (0.2)
Fetal growth restriction	9 (0.2)	7 (0.2)
Gestational diabetes	23 (0.6)	16 (0.4)
Gestational hypertension	30 (0.8)	22 (0.6)
Oligohydramnios	8 (0.2)	8 (0.2)
Polyhydramnios	8 (0.2)	11 (0.3)
Postpartum hemorrhage	5 (0.1)	7 (0.2)
Pre-eclampsia	37 (1.0)	32 (0.9)
Premature delivery	79 (2.1)	70 (1.9)
Premature labor	13 (0.4)	11 (0.3)
Premature rupture of membranes	12 (0.3)	7 (0.2)
Premature separation of placenta	6 (0.2)	4 (0.1)
Preterm premature rupture of membranes	9 (0.2)	4 (0.1)
Threatened labor	16 (0.4)	11 (0.3)
Uterine contractions during pregnancy	2 (<0.1)	7 (0.2)
Vascular disorders	13 (0.4)	14 (0.4)
Hypertension	10 (0.3)	8 (0.2)

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the maternal safety population (**Table S2**). The data cutoff date was September 02, 2022.

* Adverse events reported in $\geq 0.2\%$ of participants in either group and the associated system organ class.

† Number of participants in the specified group. This value is the denominator for the percentage calculations.

‡ Number of participants reporting ≥ 1 occurrence of the specified event category.

Table S12. Adverse events* reported in infant participants from birth through 1 month of age

System organ class Preferred term	RSVpreF (N†=3568) n‡ (%)	Placebo (N†=3558) n‡ (%)
Cardiac disorders	26 (0.7)	27 (0.8)
Bradycardia	6 (0.2)	9 (0.3)
Pulmonary valve stenosis	6 (0.2)	7 (0.2)
Congenital, familial, and genetic disorders	277 (7.8)	281 (7.9)
Ankyloglossia congenital	53 (1.5)	38 (1.1)
Atrial septal defect	30 (0.8)	46 (1.3)
Birth mark	9 (0.3)	8 (0.2)
Congenital nevus	42 (1.2)	29 (0.8)
Congenital skin dimples	7 (0.2)	10 (0.3)
Cryptochism	7 (0.2)	13 (0.4)
Developmental hip dysplasia	10 (0.3)	10 (0.3)
Hydrocele	15 (0.4)	6 (0.2)
Hypospadias	7 (0.2)	11 (0.3)
Labial tie	6 (0.2)	7 (0.2)
Laryngomalacia	8 (0.2)	2 (<0.1)
Microcephaly	5 (0.1)	7 (0.2)
Nevus flammeus	14 (0.4)	12 (0.3)
Patent ductus arteriosus	12 (0.3)	11 (0.3)
Polydactyly	6 (0.2)	9 (0.3)
Ventricular septal defect	15 (0.4)	20 (0.6)
Eye disorders	23 (0.6)	27 (0.8)
Conjunctival hemorrhage	3 (<0.1)	9 (0.3)
Dacryostenosis acquired	11 (0.3)	7 (0.2)
Gastrointestinal disorders	121 (3.4)	97 (2.7)
Constipation	13 (0.4)	15 (0.4)
Flatulence	11 (0.3)	5 (0.1)
Gastroesophageal reflux disease	24 (0.7)	19 (0.5)
Infantile colic	8 (0.2)	6 (0.2)
Umbilical hernia	21 (0.6)	19 (0.5)
Vomiting	11 (0.3)	5 (0.1)
General disorders and administration site conditions	40 (1.1)	38 (1.1)
Fever neonatal	6 (0.2)	2 (<0.1)
Hypothermia	7 (0.2)	5 (0.1)
Macrosomia	11 (0.3)	9 (0.3)
Pyrexia	10 (0.3)	12 (0.3)
Hepatobiliary disorders	112 (3.1)	105 (3.0)
Hyperbilirubinemia neonatal	108 (3.0)	104 (2.9)
Infections and infestations	122 (3.4)	125 (3.5)
Conjunctivitis	15 (0.4)	16 (0.4)
Conjunctivitis bacterial	2 (<0.1)	9 (0.3)
Infection	18 (0.5)	14 (0.4)
Oral candidiasis	9 (0.3)	18 (0.5)
Sepsis	8 (0.2)	4 (0.1)
Sepsis neonatal	20 (0.6)	21 (0.6)
Injury, poisoning, and procedural complications	33 (0.9)	43 (1.2)
Clavicle fracture	5 (0.1)	11 (0.3)
Contusion	10 (0.3)	7 (0.2)
Investigations	86 (2.4)	74 (2.1)
Apgar score low	8 (0.2)	4 (0.1)
Body temperature increased	12 (0.3)	12 (0.3)
Cardiac murmur	21 (0.6)	24 (0.7)
Oxygen saturation decreased	7 (0.2)	4 (0.1)
Metabolism and nutrition disorders	120 (3.4)	110 (3.1)
Dehydration	7 (0.2)	5 (0.1)
Hypoglycemia	42 (1.2)	39 (1.1)
Hypoglycemia neonatal	39 (1.1)	29 (0.8)
Poor feeding infant	9 (0.3)	15 (0.4)
Weight gain poor	7 (0.2)	5 (0.1)
Musculoskeletal and connective tissue disorders	26 (0.7)	23 (0.6)

Torticollis	6 (0.2)	5 (0.1)
Neoplasms benign, malignant, and unspecified (incl cysts and polyps)	21 (0.6)	24 (0.7)
Acrochordon	1 (<0.1)	7 (0.2)
Hemangioma	12 (0.3)	8 (0.2)
Nervous system disorders	40 (1.1)	25 (0.7)
Cerebral cyst	6 (0.2)	3 (<0.1)
Hypoxic-ischemic encephalopathy	7 (0.2)	3 (<0.1)
Pregnancy, puerperium, and perinatal conditions	600 (16.8)	554 (15.6)
Caput succedaneum	18 (0.5)	23 (0.6)
Cephalhematoma	15 (0.4)	16 (0.4)
Jaundice neonatal	257 (7.2)	240 (6.7)
Large for dates baby	23 (0.6)	17 (0.5)
Low birth weight baby	181 (5.1)	155 (4.4)
Meconium stain	6 (0.2)	9 (0.3)
Premature baby	202 (5.7)	169 (4.7)
Small for dates baby	32 (0.9)	38 (1.1)
Umbilical cord around neck	15 (0.4)	13 (0.4)
Umbilical granuloma	9 (0.3)	13 (0.4)
Renal and urinary disorders	15 (0.4)	26 (0.7)
Hydronephrosis	5 (0.1)	9 (0.3)
Pyelocaliectasis	5 (0.1)	8 (0.2)
Respiratory, thoracic, and mediastinal disorders	250 (7.0)	237 (6.7)
Cyanosis neonatal	10 (0.3)	12 (0.3)
Dyspnea	6 (0.2)	1 (<0.1)
Hypoxia	8 (0.2)	4 (0.1)
Infant apnea	14 (0.4)	6 (0.2)
Meconium aspiration syndrome	12 (0.3)	8 (0.2)
Neonatal asphyxia	9 (0.3)	9 (0.3)
Neonatal respiratory distress	19 (0.5)	20 (0.6)
Neonatal respiratory distress syndrome	11 (0.3)	15 (0.4)
Pneumothorax	7 (0.2)	6 (0.2)
Respiratory disorder neonatal	6 (0.2)	4 (0.1)
Respiratory distress	67 (1.9)	63 (1.8)
Tachypnea	14 (0.4)	20 (0.6)
Transient tachypnea of the newborn	47 (1.3)	48 (1.3)
Skin and subcutaneous tissue disorders	134 (3.8)	123 (3.5)
Dermatitis diaper	30 (0.8)	36 (1.0)
Eczema infantile	9 (0.3)	12 (0.3)
Erythema toxicum neonatorum	16 (0.4)	12 (0.3)
Rash	19 (0.5)	17 (0.5)
Seborrheic dermatitis	12 (0.3)	5 (0.1)
Vascular disorders	10 (0.3)	15 (0.4)
Cyanosis	4 (0.1)	8 (0.2)

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the infant safety population (**Table S2**). The data cutoff date was September 02, 2022.

* Adverse events reported in $\geq 0.2\%$ of participants in either group and the associated system organ class.

† Number of participants in the specified group. This value is the denominator for the percentage calculations.

‡ Number of participants reporting ≥ 1 occurrence of the specified event category.

Table S13. Serious adverse events* reported in maternal participants from vaccination through 6 months after delivery

System organ class	RSVpreF (N†=3682)	Placebo (N†=3675)
Preferred term	n‡ (%)	n‡ (%)
Cardiac disorders	60 (1.6)	60 (1.6)
Bradycardia fetal	8 (0.2)	12 (0.3)
Fetal heartrate deceleration abnormality	6 (0.2)	12 (0.3)
Nonreassuring fetal heartrate pattern	36 (1.0)	29 (0.8)
Hepatobiliary disorders	17 (0.5)	17 (0.5)
Cholestasis of pregnancy	11 (0.3)	6 (0.2)
Infections and infestations	47 (1.3)	44 (1.2)
Amniotic cavity infection	8 (0.2)	5 (0.1)
Urinary tract infection	6 (0.2)	6 (0.2)
Investigations	21 (0.6)	15 (0.4)
Fetal heart rate abnormal	7 (0.2)	2 (<0.1)
Fetal monitoring abnormal	4 (0.1)	6 (0.2)
Pregnancy, puerperium, and perinatal conditions	446 (12.1)	411 (11.2)
Abnormal labor	5 (0.1)	7 (0.2)
Arrested labor	36 (1.0)	42 (1.1)
Breech presentation	6 (0.2)	5 (0.1)
Cephalopelvic disproportion	14 (0.4)	19 (0.5)
Failed induction of labor	8 (0.2)	8 (0.2)
False labor	2 (<0.1)	6 (0.2)
Fetal death/stillborn baby	10 (0.3)	8 (0.2)
Fetal distress syndrome	66 (1.8)	60 (1.6)
Fetal growth restriction	15 (0.4)	10 (0.3)
Fetal macrosomia	5 (0.1)	8 (0.2)
Gestational hypertension	41 (1.1)	38 (1.0)
Hemorrhage in pregnancy	7 (0.2)	3 (<0.1)
Obstructed labor	15 (0.4)	11 (0.3)
Oligohydramnios	15 (0.4)	20 (0.5)
Postpartum hemorrhage	28 (0.8)	32 (0.9)
Preelampsia	68 (1.8)	53 (1.4)
Premature delivery	28 (0.8)	23 (0.6)
Premature labor	16 (0.4)	10 (0.3)
Premature rupture of membranes	15 (0.4)	16 (0.4)
Premature separation of placenta	19 (0.5)	12 (0.3)
Preterm premature rupture of membranes	15 (0.4)	10 (0.3)
Prolonged labor	23 (0.6)	21 (0.6)
Prolonged pregnancy	9 (0.2)	7 (0.2)
Prolonged rupture of membranes	9 (0.2)	10 (0.3)
Retained placenta or membranes	8 (0.2)	5 (0.1)
Threatened labor	13 (0.4)	14 (0.4)
Vascular disorders	14 (0.4)	13 (0.4)
Hypertension	13 (0.4)	6 (0.2)

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the maternal safety population (**Table S2**). The data cutoff date was September 02, 2022.

* Serious adverse events reported in $\geq 0.2\%$ of participants in either group and the associated system organ class.

† Number of participants in the specified group. This value is the denominator for the percentage calculations.

‡ Number of participants reporting ≥ 1 occurrence of the specified event category.

Table S14. Serious adverse events* reported in infant participants from birth through 24 months of age

System organ class Preferred term	RSVpreF (N†=3568) n‡ (%)	Placebo (N†=3558) n‡ (%)
Cardiac disorders	19 (0.5)	14 (0.4)
Pulmonary valve stenosis	6 (0.2)	6 (0.2)
Congenital, familial, and genetic disorders	152 (4.3)	189 (5.3)
Ankyloglossia congenital	15 (0.4)	10 (0.3)
Atrial septal defect	31 (0.9)	40 (1.1)
Cryptochism	7 (0.2)	16 (0.4)
Developmental hip dysplasia	11 (0.3)	12 (0.3)
Hypospadias	7 (0.2)	11 (0.3)
Microcephaly	5 (0.1)	7 (0.2)
Patent ductus arteriosus	12 (0.3)	10 (0.3)
Ventricular septal defect	15 (0.4)	20 (0.6)
Gastrointestinal disorders	24 (0.7)	29 (0.8)
Inguinal hernia	6 (0.2)	6 (0.2)
General disorders and administration site conditions	13 (0.4)	19 (0.5)
Pyrexia	2 (<0.1)	6 (0.2)
Hepatobiliary disorders	53 (1.5)	42 (1.2)
Hyperbilirubinemia neonatal	49 (1.4)	40 (1.1)
Infections and infestations	108 (3.0)	90 (12.5)
Gastroenteritis	14 (0.4)	12 (0.3)
Infection	13 (0.4)	12 (0.3)
Sepsis	9 (0.3)	4 (0.1)
Sepsis neonatal	19 (0.5)	19 (0.5)
Urinary tract infection	12 (0.3)	14 (0.4)
Metabolism and nutrition disorders	51 (1.4)	44 (1.2)
Dehydration	6 (0.2)	8 (0.2)
Failure to thrive	6 (0.2)	5 (0.1)
Hypoglycemia	20 (0.6)	17 (0.5)
Hypoglycemia neonatal	13 (0.4)	10 (0.3)
Musculoskeletal and connective tissue disorders	17 (0.5)	13 (0.4)
Torticollis	5 (0.1)	7 (0.2)
Nervous system disorders	42 (1.2)	31 (0.9)
Cerebral cyst	6 (0.2)	3 (<0.1)
Febrile convulsion	5 (0.1)	6 (0.2)
Hypoxic-ischemic encephalopathy	6 (0.2)	3 (<0.1)
Pregnancy, puerperium, and perinatal conditions	140 (3.9)	126 (3.5)
Jaundice neonatal	75 (2.1)	66 (1.9)
Low birth weight baby	27 (0.8)	31 (0.9)
Premature baby	49 (1.4)	42 (1.2)
Small for dates baby	6 (0.2)	10 (0.3)
Renal and urinary disorders	12 (0.3)	22 (0.6)
Hydronephrosis	5 (0.1)	9 (0.3)
Pyelocaliectasis	5 (0.1)	8 (0.2)

Respiratory, thoracic and mediastinal disorders	163 (4.6)	149 (4.2)
Hypoxia	6 (0.2)	3 (<0.1)
Infantile apnea	10 (0.3)	3 (<0.1)
Meconium aspiration syndrome	9 (0.3)	7 (0.2)
Neonatal asphyxia	8 (0.2)	7 (0.2)
Neonatal respiratory distress	11 (0.3)	14 (0.4)
Neonatal respiratory distress syndrome	10 (0.3)	14 (0.4)
Respiratory distress	47 (1.3)	43 (1.2)
Tachypnea	6 (0.2)	8 (0.2)
Transient tachypnea of the newborn	33 (0.9)	29 (0.8)

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the infant safety population (**Table S2**). The data cutoff date was September 02, 2022.

* Serious adverse events reported in $\geq 0.2\%$ of participants in either group and the associated system organ class.

† Number of participants in the specified group. This value is the denominator for the percentage calculations.

‡ Number of participants reporting ≥ 1 occurrence of the specified event category.

Table S15. Causes of death in infant participants*

Time interval	Term	RSVpreF (N†=3568) n‡ (%)	Placebo (N†=3558) n‡ (%)
Birth to 24 months of age	Any death event	5 (0.1)	12 (0.3)
	Death	0	1 (<0.1)§
	Enterovirus infection	1 (<0.1)	0
	Gastroenteritis	1 (<0.1)	0
	Ill-defined disorder	1 (<0.1)	1 (<0.1)
	Interstitial lung disease	1 (<0.1)	0
	Low birth weight baby	0	1 (<0.1)
	Meconium aspiration syndrome	1 (<0.1)	1 (<0.1)
	Meningitis	0	1 (<0.1)
	Multiple organ dysfunction syndrome	0	1 (<0.1)
	Neonatal asphyxia	0	2 (<0.1)
	Neonatal respiratory distress	0	1 (<0.1)
	Pneumonia	0	1 (<0.1)
	Premature baby	1 (<0.1)	1 (<0.1)
	Pulmonary hypertension	0	1 (<0.1)
	Rhinovirus infection	1 (<0.1)	0
	Sepsis neonatal	0	2 (<0.1)
	Small for dates baby	0	1 (<0.1)
	Sudden infant death syndrome	0	3 (<0.1)
	Ventricular hypoplasia	0	1 (<0.1)
Birth to 1 month of age	Any death event	2 (<0.1)	5 (0.1)
	Meconium aspiration syndrome	1 (<0.1)	1 (<0.1)
	Multiple organ dysfunction syndrome	0	1 (<0.1)
	Neonatal asphyxia	0	2 (<0.1)
	Premature baby	1 (<0.1)	0
	Pulmonary hypertension	0	1 (<0.1)
	Sepsis neonatal	0	2 (<0.1)
1–6 months of age	Ventricular hypoplasia	0	1 (<0.1)
	Any death event	3 (<0.1)	6 (0.2)
	Death	0	1 (<0.1)
	Enterovirus infection	1 (<0.1)	0
	Gastroenteritis	1 (<0.1)	0
	Ill-defined disorder	1 (<0.1)	1 (<0.1)
	Interstitial lung disease	1 (<0.1)	0
	Low birth weight baby	0	1 (<0.1)
	Meningitis	0	1 (<0.1)
	Neonatal respiratory distress	0	1 (<0.1)
	Pneumonia	0	1 (<0.1)
	Premature baby	0	1 (<0.1)
	Rhinovirus infection	1 (<0.1)	0
	Small for dates baby	0	1 (<0.1)
	Sudden infant death syndrome	0	2 (<0.1)
6–12 months of age	Any death event	0	1 (<0.1)

Sudden infant death syndrome	0	1 (<0.1)
RSV=respiratory syncytial virus; RSVpreF=respiratory syncytial virus prefusion F protein–based vaccine. Results are for the infant safety population (Table S2). The data cutoff date was September 02, 2022. * Stillbirths were classified as serious adverse events in maternal participants and are summarized in Table S7 . † Number of participants in the vaccine group. This value is the denominator for the percentage calculations. ‡ Number of participants reporting ≥ 1 occurrence of the specified death term. For "any death event", this is the number of participants reporting ≥ 1 occurrence of any death term. § The endpoint adjudication committee considered the cause of death to be “acute respiratory illness due to RSV”. The event occurred 120 days after birth.		

Table S16. Severity scale for local reactions and systemic events (maternal participants)

	Mild Grade 1	Moderate Grade 2	Severe Grade 3	Grade 4*
Local reaction				
Redness†	>2.0–5.0 cm (5–10 measuring device units)	>5.0–10.0 cm (11–20 measuring device units)	>10 cm (>20 measuring device units)	Necrosis or exfoliative dermatitis
Swelling†	>2.0–5.0 cm (5–10 measuring device units)	>5.0–10.0 cm (11–20 measuring device units)	>10 cm (>20 measuring device units)	Necrosis
Injection site pain	Does not interfere with activity	Interferes with activity	Prevents daily activity	ER visit or hospitalization for severe pain at the injection site
Systemic event				
Fatigue	Does not interfere with activity	Some interference with activity	Prevents daily routine activity	ER visit or hospitalization for severe fatigue
Headache	Does not interfere with activity	Some interference with activity	Prevents daily routine activity	ER visit or hospitalization for severe headache
Vomiting	1–2 times in 24 hours	>2 times in 24 hours	Requires IV hydration	ER visit or hospitalization for severe vomiting
Nausea	Does not interfere with activity	Some interference with activity	Prevents daily routine activity	ER visit or hospitalization for severe nausea
Diarrhea	2–3 loose stools in 24 hours	4–5 loose stools in 24 hours	≥6 loose stools in 24 hours	ER visit or hospitalization for severe diarrhea
Muscle pain	Does not interfere with activity	Some interference with activity	Prevents daily routine activity	ER visit or hospitalization for severe muscle pain
Joint pain	Does not interfere with activity	Some interference with activity	Prevents daily routine activity	ER visit or hospitalization for severe joint pain
Fever	38.0°C–38.4°C	>38.4°C–38.9°C	>38.9°C–40.0°C	>40.0°C

AE=adverse event; ER=emergency room; IV=intravenous.

* Only an investigator or qualified designee was able to classify a participant's local reaction as grade 4, after clinical evaluation of the participant or documentation from another medically qualified source (eg, ER or hospital record) or, in the case of pain at the injection site only, contact with the participant. Grade 4 local reactions and systemic events (except fever) were collected on the AE case report form and assessed by the investigator using the AE intensity grading scale in the table above.

† Measured by the maternal participant, study staff or field worker and recorded in measuring device units (range, 1–20 and ≥21) and then categorized during analysis as mild, moderate, or severe based on the grading scale in the table. Measuring device units could be converted to centimeters according to the following scale: 1 measuring device unit=0.5 cm.

Table S17. Severe or life-threatening adverse events reported in maternal participants within 1 month of vaccination

System organ class Preferred term	RSVpreF (N*=3682) n† (%)	Placebo (N*=3675) n† (%)
Any	81 (2.2)	56 (1.5)
Blood and lymphatic disorders	0	1 (<0.1)
Anemia	0	1 (<0.1)
Cardiac disorders	6 (0.2)	3 (<0.1)
Bradycardia fetal	0	2 (<0.1)
Fetal heart rate deceleration abnormality	1 (<0.1)	1 (<0.1)
Mitral valve incompetence	1 (<0.1)	0
Nonreassuring fetal heart rate pattern	3 (<0.1)	0
Tachycardia	1 (<0.1)	0
Gastrointestinal disorders	0	3 (<0.1)
Abdominal pain	0	1 (<0.1)
Diarrhea	0	1 (<0.1)
Hemorrhoidal hemorrhage	0	1 (<0.1)
Hepatobiliary disorders	2 (<0.1)	5 (0.1)
Cholestasis	0	1 (<0.1)
Cholestasis of pregnancy	2 (<0.1)	4 (0.1)
Immune system disorders	1 (<0.1)	0
Rhesus incompatibility	1 (<0.1)	0
Infections and infestations	6 (0.2)	3 (<0.1)
Endometritis	1 (<0.1)	0
Gastroenteritis	2 (<0.1)	0
Pilonidal disease	0	1 (<0.1)
Pyelonephritis	0	1 (<0.1)
Urinary tract infection	3 (<0.1)	1 (<0.1)
Injury, poisoning, and procedural complications	3 (<0.1)	1 (<0.1)
Ankle fracture	1 (<0.1)	0
Perineal injury	1 (<0.1)	0
Uterine rupture	1 (<0.1)	0
Vulvovaginal injury	0	1 (<0.1)
Investigations	2 (<0.1)	0
Body temperature increased	1 (<0.1)	0
Fetal heart rate abnormal	1 (<0.1)	0
Metabolism and nutrition disorders	1 (<0.1)	0
Diabetes mellitus	1 (<0.1)	0
Musculoskeletal and connective tissue disorders	1 (<0.1)	0
Pain in extremity	1 (<0.1)	0
Nervous system disorders	0	1 (<0.1)
Epilepsy	0	1 (<0.1)
Pregnancy, puerperium, and perinatal conditions	63 (1.7)	36 (1.0)
Abnormal labor	1 (<0.1)	0
Arrested labor	2 (<0.1)	1 (<0.1)
Cephalo-pelvic disproportion	0	1 (<0.1)
Eclampsia	1 (<0.1)	2 (<0.1)
False labor	0	1 (<0.1)
Fetal death‡	2 (<0.1)	2 (<0.1)
Fetal distress syndrome	4 (0.1)	5 (0.1)
Fetal growth restriction	1 (<0.1)	1 (<0.1)
Fetal hypokinesia	2 (<0.1)	0
Fetal vascular malperfusion	0	1 (<0.1)
Fetal-maternal hemorrhage	1 (<0.1)	0
Gestational hypertension	2 (<0.1)	1 (<0.1)
HELLP syndrome	2 (<0.1)	1 (<0.1)
Hemorrhage in pregnancy	1 (<0.1)	2 (<0.1)
Obstructed labour	0	1 (<0.1)
Oligohydramnios	2 (<0.1)	1 (<0.1)
Placental insufficiency	1 (<0.1)	0
Polyhydramnios	0	1 (<0.1)
Postpartum hemorrhage	2 (<0.1)	0
Pre-eclampsia	17 (0.5)	7 (0.2)

Premature delivery	8 (0.2)	4 (0.1)
Premature labor	4 (0.1)	0
Premature rupture of membranes	3 (<0.1)	2 (<0.1)
Premature separation of placenta	2 (<0.1)	3 (<0.1)
Preterm premature rupture of membranes	3 (<0.1)	0
Prolonged rupture of membranes	0	1 (<0.1)
Retained placenta or membranes	2 (<0.1)	0
Spontaneous rupture of membranes	0	1 (<0.1)
Threatened labor	4 (0.1)	3 (<0.1)
Umbilical cord abnormality	1 (<0.1)	1 (<0.1)
Umbilical cord prolapse	0	2 (<0.1)
Uterine atony	2 (<0.1)	0
Psychiatric disorders	1 (<0.1)	0
Anxiety disorder	1 (<0.1)	0
Renal and urinary disorders	2 (<0.1)	0
Nephrolithiasis	1 (<0.1)	0
Renal impairment	1 (<0.1)	0
Reproductive system and breast disorders	1 (<0.1)	1 (<0.1)
Uterine hemorrhage	1 (<0.1)	0
Vaginal discharge	0	1 (<0.1)
Skin and subcutaneous tissue disorders	0	1 (<0.1)
Dermal cyst	0	1 (<0.1)
Vascular disorders	0	3 (<0.1)
Hypertension	0	2 (<0.1)
Hypotension	0	1 (<0.1)

HELLP=hemolysis, elevated liver enzymes, and low platelets; RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the maternal safety population (**Table S2**). The data cutoff date was September 02, 2022.

* Number of participants in the specified group. This value is the denominator for the percentage calculations.

† Number of participants reporting ≥ 1 occurrence of the specified event category.

‡ This event refers to the maternal participant having a stillborn baby.

Table S18. Severe or life-threatening adverse events reported in infant participants from birth to 1 month of age

System organ class Preferred term	RSVpreF (N*=3568) n† (%)	Placebo (N*=3558) n† (%)
Any	183 (5.1)	160 (4.5)
Blood and lymphatic disorders	5 (0.1)	2 (<0.1)
Anemia neonatal	2 (<0.1)	0
Hemolysis neonatal	1 (<0.1)	1 (<0.1)
Lymphadenitis	0	1 (<0.1)
Polycythemia	1 (<0.1)	0
Polycythemia neonatorum	1 (<0.1)	0
Thrombocytopenia neonatal	1 (<0.1)	0
Cardiac disorders	6 (0.2)	3 (<0.1)
Aortic valve incompetence	1 (<0.1)	0
Atrioventricular block complete	1 (<0.1)	0
Bradycardia	2 (<0.1)	2 (<0.1)
Bradycardia neonatal	1 (<0.1)	0
Mitral valve incompetence	1 (<0.1)	0
Pulmonary valve stenosis	0	1 (<0.1)
Pulmonary valve thickening	1 (<0.1)	0
Tricuspid valve incompetence	1 (<0.1)	0
Congenital, familial, and genetic disorders	22 (0.6)	28 (0.8)
Ankyloglossia congenital	3 (<0.1)	0
Atrial septal defect	3 (<0.1)	5 (0.1)
Autosomal chromosome anomaly	1 (<0.1)	0
Chromosomal deletion	0	1 (<0.1)
Cleft lip and palate	0	1 (<0.1)
Cleft palate	0	1 (<0.1)
Coarctation of the aorta	0	1 (<0.1)
Congenital central nervous system anomaly	1 (<0.1)	0
Congenital cytomegalovirus infection	0	1 (<0.1)
Congenital foot malformation	0	1 (<0.1)
Congenital megacolon	1 (<0.1)	0
Congenital monorchidism	1 (<0.1)	0
Congenital syphilis	2 (<0.1)	0
Craniosynostosis	1 (<0.1)	0
Cystic fibrosis	1 (<0.1)	0
Deafness congenital	0	1 (<0.1)
Developmental hip dysplasia	2 (<0.1)	2 (<0.1)
Galactosemia	0	1 (<0.1)
Genitalia external ambiguous	0	1 (<0.1)
Heart disease congenital	0	1 (<0.1)
Hypoplastic left heart syndrome	0	1 (<0.1)
Hypospadias	0	1 (<0.1)
Inborn error of metabolism	1 (<0.1)	0
Intestinal malrotation	0	1 (<0.1)
Labial tie	0	1 (<0.1)
Laryngomalacia	0	1 (<0.1)
Left-to-right cardiac shunt	1 (<0.1)	1 (<0.1)
Newborn persistent pulmonary hypertension	0	3 (<0.1)
Patent ductus arteriosus	1 (<0.1)	1 (<0.1)
Pyloric stenosis	1 (<0.1)	1 (<0.1)
Renal dysplasia	1 (<0.1)	0
Spina bifida occulta	0	1 (<0.1)
Talipes	2 (<0.1)	1 (<0.1)
Ventricular hypoplasia	0	1 (<0.1)
Ventricular septal defect	2 (<0.1)	3 (<0.1)
Gastrointestinal disorders	5 (0.1)	4 (0.1)
Constipation	0	1 (<0.1)
Diarrhea	0	1 (<0.1)
Gastroesophageal reflux disease	0	2 (<0.1)
Ileus	1 (<0.1)	0
Infantile colic	1 (<0.1)	0

Meconium ileus	1 (<0.1)	0
Necrotizing colitis	2 (<0.1)	0
Obstruction gastric	1 (<0.1)	0
Pneumoperitoneum	1 (<0.1)	0
Vomiting	0	1 (<0.1)
General disorders and administration site conditions	3 (<0.1)	1 (<0.1)
Drug withdrawal syndrome neonatal	1 (<0.1)	0
Hypothermia	2 (<0.1)	0
Multiple organ dysfunction syndrome	0	1 (<0.1)
Hepatobiliary disorders	7 (0.2)	10 (0.3)
Hepatitis	0	1 (<0.1)
Hyperbilirubinemia neonatal	7 (0.2)	9 (0.3)
Immune system disorders	0	2 (<0.1)
ABO incompatibility	0	2 (<0.1)
Infections and infestations	30 (0.8)	21 (0.6)
Bacterial infection	1 (<0.1)	0
Bacterial sepsis	2 (<0.1)	0
Cellulitis staphylococcal	0	1 (<0.1)
Gastroenteritis	0	1 (<0.1)
Groin abscess	0	1 (<0.1)
Infection	6 (0.2)	0
Meningitis bacterial	0	2 (<0.1)
Meningitis enteroviral	0	1 (<0.1)
Meningitis streptococcal	1 (<0.1)	0
Meningitis viral	1 (<0.1)	0
Neonatal infection	0	2 (<0.1)
Neonatal infective mastitis	0	1 (<0.1)
Neonatal pneumonia	2 (<0.1)	0
Omphalitis	1 (<0.1)	0
Ophthalmia neonatorum	1 (<0.1)	0
Pneumonia	1 (<0.1)	1 (<0.1)
Pyelonephritis	1 (<0.1)	1 (<0.1)
Sepsis	5 (0.1)	4 (0.1)
Sepsis neonatal	6 (0.2)	8 (0.2)
Septic coagulopathy	1 (<0.1)	0
Urinary tract infection	2 (<0.1)	1 (<0.1)
Injury, poisoning, and procedural complications	3 (<0.1)	4 (0.1)
Clavicle fracture	0	1 (<0.1)
Head injury	1 (<0.1)	0
Humerus fracture	0	1 (<0.1)
Injury to brachial plexus due to birth trauma	0	1 (<0.1)
Maternal drugs affecting fetus	1 (<0.1)	0
Skull fracture	0	1 (<0.1)
Subdural hematoma	1 (<0.1)	0
Investigations	2 (<0.1)	1 (<0.1)
Apgar score low	1 (<0.1)	1 (<0.1)
Nutritional condition abnormal	1 (<0.1)	0
Metabolism and nutrition disorders	23 (0.6)	18 (0.5)
Acidosis	0	1 (<0.1)
Dehydration	1 (<0.1)	2 (<0.1)
Failure to thrive	1 (<0.1)	1 (<0.1)
Feeding disorder	1 (<0.1)	0
Feeding intolerance	1 (<0.1)	1 (<0.1)
Hyperkalemia	1 (<0.1)	0
Hypernatremia	1 (<0.1)	0
Hypocalcemia	0	1 (<0.1)
Hypoglycemia	10 (0.3)	10 (0.3)
Hypoglycemia neonatal	6 (0.2)	2 (<0.1)
Lactic acidosis	1 (<0.1)	0
Metabolic acidosis	2 (<0.1)	1 (<0.1)
Neonatal hyponatremia	1 (<0.1)	0
Poor feeding infant	0	1 (<0.1)
Musculoskeletal and connective tissue disorders	1 (<0.1)	2 (<0.1)
Hypotonia neonatal	1 (<0.1)	0
Torticollis	0	2 (<0.1)
Nervous system disorders	10 (0.3)	4 (0.1)
Encephalopathy neonatal	1 (<0.1)	2 (<0.1)

Hyporeflexia	1 (<0.1)	0
Hypoxic-ischemic encephalopathy	5 (0.1)	0
Intracranial hemorrhage neonatal	1 (<0.1)	0
Intraventricular hemorrhage	1 (<0.1)	0
Intraventricular hemorrhage neonatal	0	1 (<0.1)
Seizure	1 (<0.1)	0
Subarachnoid hemorrhage	1 (<0.1)	0
Subdural hemorrhage neonatal	0	1 (<0.1)
Pregnancy, puerperium, and perinatal conditions	41 (1.1)	33 (0.9)
Birth trauma	0	1 (<0.1)
Fetal distress syndrome	1 (<0.1)	0
Jaundice neonatal	21 (0.6)	11 (0.3)
Low birth weight baby	9 (0.3)	6 (0.2)
Meconium stain	0	2 (<0.1)
Neonatal disorder	2 (<0.1)	1 (<0.1)
Premature baby	16 (0.4)	14 (0.4)
Small for dates baby	1 (<0.1)	3 (<0.1)
Umbilical cord around neck	1 (<0.1)	0
Renal and urinary disorders	0	3 (<0.1)
Hydronephrosis	0	2 (<0.1)
Hydroureter	0	1 (<0.1)
Vesicoureteric reflux	0	1 (<0.1)
Reproductive system and breast disorders	1 (<0.1)	0
Testicular torsion	1 (<0.1)	0
Respiratory, thoracic, and mediastinal disorders	87 (2.4)	67 (1.9)
Acute respiratory distress syndrome	1 (<0.1)	0
Apnea	1 (<0.1)	2 (<0.1)
Asphyxia	1 (<0.1)	0
Brief resolved unexplained event	0	1 (<0.1)
Choking	1 (<0.1)	0
Dyspnea	2 (<0.1)	0
Hypoxia	5 (0.1)	1 (<0.1)
Infantile apnea	4 (0.1)	3 (<0.1)
Meconium aspiration syndrome	5 (0.1)	3 (<0.1)
Nasal flaring	1 (<0.1)	0
Neonatal asphyxia	4 (0.1)	7 (0.2)
Neonatal aspiration	0	1 (<0.1)
Neonatal dyspnea	1 (<0.1)	0
Neonatal hypoxia	0	1 (<0.1)
Neonatal respiratory depression	1 (<0.1)	1 (<0.1)
Neonatal respiratory distress	8 (0.2)	7 (0.2)
Neonatal respiratory distress syndrome	4 (0.1)	7 (0.2)
Neonatal respiratory failure	1 (<0.1)	3 (<0.1)
Neonatal tachypnoea	1 (<0.1)	0
Pneumonitis aspiration	0	1 (<0.1)
Pneumothorax	2 (<0.1)	0
Pulmonary arterial hypertension	1 (<0.1)	0
Pulmonary hypertension	0	2 (<0.1)
Respiratory depression	1 (<0.1)	1 (<0.1)
Respiratory disorder neonatal	1 (<0.1)	0
Respiratory distress	30 (0.8)	21 (0.6)
Respiratory failure	1 (<0.1)	2 (<0.1)
Tachypnea	3 (<0.1)	0
Transient tachypnea of the newborn	17 (0.5)	11 (0.3)
Skin and subcutaneous tissue disorders	1 (<0.1)	0
Skin discolouration	1 (<0.1)	0
Surgical and medical procedures	0	1 (<0.1)
Infection prophylaxis	0	1 (<0.1)
Vascular disorders	0	2 (<0.1)
Cyanosis	0	1 (<0.1)
Subgaleal hematoma	0	1 (<0.1)

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the maternal safety population (**Table S2**). The data cutoff date was September 02, 2022.

* Number of participants in the specified group. This value is the denominator for the percentage calculations.

† Number of participants reporting ≥ 1 occurrence of the specified event category.

Table S19. Time from vaccination to delivery among preterm deliveries

	RSVpreF (N*=3577) n† (%)	Placebo (N*=3570) n† (%)	Total (N*=7147) n† (%)
Number of preterm deliveries‡	229	198	427
Time from vaccination to delivery			
≤7 days	11 (4.8)	16 (8.1)	27 (6.3)
>7 days to ≤30 days	81 (35.4)	71 (35.9)	152 (35.6)
>30 days	137 (59.8)	111 (56.1)	248 (58.1)

RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

Results are for the maternal safety population (**Table S2**).

One participant had a missing delivery date in the database, so is not included in counts above; the participant's baby was born at term, more than 30 days after vaccination.

Preterm deliveries were determined based on gestational age at vaccination and the time between vaccination and delivery. Preterm was defined as gestational age at delivery <37 weeks. The number of days between vaccination and delivery was calculated as delivery date minus vaccination date.

* Number of participants having delivery date in the specified vaccine group.

† Number of participants in the specified category.

‡ This value is the denominator for the percentage calculations.

Figure S1. Newly diagnosed chronic medical conditions* in infant participants from birth through 24 months of age

* Reported in $\geq 0.2\%$ of participants in either group. Results are for the infant safety population (Table S2). The data cutoff date was September 02, 2022. Exact two-sided 95% CIs were calculated using the Clopper and Pearson method. GERD=gastroesophageal reflux disease; RSVpreF=respiratory syncytial virus prefusion F protein-based vaccine.

